

# Spell Correction

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آکادمی هراه



- **Introduction**
- **Definition of Minimum Edit Distance**
- **Computing Minimum Edit Distance**
- **Backtrace for Computing Alignments**



## Who cares about spelling?

According to a research at Cambridge University, it doesn't matter in what order the letters in a word are, the only important thing is that the first and last letter be at the right place. The rest can be a total mess and you can still read it without problem. This is because the human mind does not read every letter by itself, but the word as a whole.

**(See for the story behind this supposed research report.)**



## Detection vs. Correction

➤ Two distinct tasks:

▶ Error detection = simply find the misspelled words

▶ Error correction = correct the misspelled words

➤ It might be easy to tell that "ater" is a misspelled word, but what is the correct word?

water? later? after?



## What causes errors?

- Keyboard mistyping Space bar
- Keyboard proximity
- Similarity of shape
- Phonetic errors
- Knowledge problems



## Keyboard Space Bar and Keyboard Proximity

### ➤ Space bar issues

- ▶ Run-on errors = two separate words become one
  - ▶ e.g., the fuzz becomes thefuzz
- ▶ Split errors = one word becomes two separate words
  - ▶ e.g., equalization becomes equali zation

### ➤ Keyboard proximity

- ▶ e.g., Jack becomes Hack since h, j are next to each other on a typical American keyboard



## Similarity of Shape

### ➤ Physical similarity

- ▶ Similarity of shape, e.g., mistaking two physically similar letters when typing up something handwritten
  - ▶ e.g., tight for fight



## Phonetic Errors

### ➤ Errors based on the sounds of a language (not necessarily on the letters)

▶ Homophones = two words which sound the same

▶ e.g., red/ read (past tense), cite/ site/ sight, they're/ their/ there

▶ Spoonerisms = switching two letters/sounds around

▶ e.g., It's a ravy(vary) grain with biscuit wheels.

▶ Letter substitution = replacing a letter (or sequence of letters) with a similar-sounding one

▶ e.g., John kracked his nuckles. instead of John cracked his knuckles

▶ e.g., I study sikologee





## Knowledge Problems

### ➤ Cases of not knowing how to spell:

▶ Not knowing a word and guessing its spelling (can be phonetic)

▶ e.g., sientist

▶ Not knowing a rule and guessing it

▶ e.g., Do we double a consonant for ing words? jog → jogging ?



## Spell Checking Approaches

- Minimum edit distance
- Language Modeling
- Sequence to Sequence Models
- ....



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## How similar are two strings?

### ➤ Spell correction

- ▶ The user typed "graffe"

Which is closest?

- ▶ graf
- ▶ graft
- ▶ grail
- ▶ giraffe

### ➤ Computational Biology

- ▶ Align two sequences of nucleotides

```
AGGCTATCACCTGACCTCCAGGCCGATGCCC  
TAGCTATCACGACCGCGGTCGATTGCCCCGAC
```

- ▶ Resulting alignment:

```
-AGGCTATCACCTGACCTCCAGGCCGA--TGCCC---  
TAG-CTATCAC--GACCGC--GGTCGATTGCCCCGAC
```

### ➤ Also for Machine Translation, Information Extraction, Speech Recognition



➤ The minimum edit distance between two strings is the minimum number of editing operations

- ▶ Insertion
- ▶ Deletion
- ▶ Substitution

➤ Needed to transform one into the other



## Minimum Edit Distance

➤ Two strings and their **alignment**:

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| I | N | T | E | * | N | T | I | O | N |
|   |   |   |   |   |   |   |   |   |   |
| * | E | X | E | C | U | T | I | O | N |



## Minimum Edit Distance

I N T E \* N T I O N  
| | | | | | | | | |  
\* E X E C U T I O N  
d s s i s

- If each operation has cost of 1
  - ▶ Distance between these is 5
- If substitutions cost 2 (Levenshtein)
  - ▶ Distance between them is 8



## Alignment in Computational Biology

➤ Given a sequence of bases

```
AGGCTATCACCTGACCTCCAGGCCGATGCCC  
TAGCTATCACGACCGCGGTCGATTGCCCCGAC
```

➤ An alignment:

```
-AGGCTATCACCTGACCTCCAGGCCGA--TGCCC---  
TAG-CTATCAC--GACCGC--GGTCGATTGCCCCGAC
```

➤ Given two sequences, align each letter to a letter or gap





## Other uses of Edit Distance in NLP

### ➤ Evaluating Machine Translation and speech recognition

**R** Spokesman confirms      senior government adviser was appointed

**H** Spokesman said      the senior      adviser was appointed

S

I

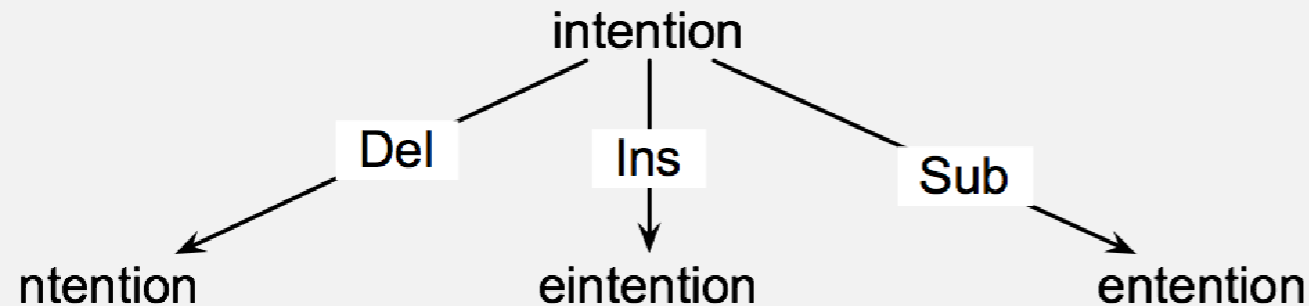
D



## How to find the Min Edit Distance?

➤ Searching for a path (sequence of edits) from the start string to the final string:

- ▶ **Initial state:** the word we're transforming
- ▶ **Operators:** insert, delete, substitute
- ▶ **Goal state:** the word we're trying to get to
- ▶ **Path cost:** what we want to minimize: the number of edits





## Minimum Edit as Search

- But the space of all edit sequences is huge!
  - ▶ We can't afford to navigate naïvely
  - ▶ Lots of distinct paths wind up at the same state.
    - ▶ We don't have to keep track of all of them
    - ▶ Just the shortest path to each of those revisited states.



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## Dynamic Programming for Minimum Edit Distance

- Dynamic programming: A tabular computation of  $D(n,m)$
- Solving problems by combining solutions to subproblems.
- Bottom-up
  - ▶ We compute  $D(i,j)$  for small  $i,j$
  - ▶ And compute larger  $D(i,j)$  based on previously computed smaller values
  - ▶ i.e., compute  $D(i,j)$  for all  $i$  ( $0 < i < n$ ) and  $j$  ( $0 < j < m$ )



## Defining Min Edit Distance (Levenshtein)

### Initialization

$$D(i,0) = i$$

$$D(0,j) = j$$

### Recurrence Relation:

For each  $i = 1 \dots M$

$$\text{For each } j = 1 \dots N \quad D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } X(i) \neq Y(j) \\ 0; & \text{if } X(i) = Y(j) \end{cases} \end{cases}$$

### Termination:

$D(N,M)$  is distance



## The Edit Distance Table

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| N | 9 |   |   |   |   |   |   |   |   |   |
| O | 8 |   |   |   |   |   |   |   |   |   |
| I | 7 |   |   |   |   |   |   |   |   |   |
| T | 6 |   |   |   |   |   |   |   |   |   |
| N | 5 |   |   |   |   |   |   |   |   |   |
| E | 4 |   |   |   |   |   |   |   |   |   |
| T | 3 |   |   |   |   |   |   |   |   |   |
| N | 2 |   |   |   |   |   |   |   |   |   |
| I | 1 |   |   |   |   |   |   |   |   |   |
| # | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|   | # | E | X | E | C | U | T | I | O | N |



## The Edit Distance Table

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| N | 9 |   |   |   |   |   |   |   |   |   |
| O | 8 |   |   |   |   |   |   |   |   |   |
| I | 7 |   |   |   |   |   |   |   |   |   |
| T | 6 |   |   |   |   |   |   |   |   |   |
| N | 5 |   |   |   |   |   |   |   |   |   |
| E | 4 |   |   |   |   |   |   |   |   |   |
| T | 3 |   |   |   |   |   |   |   |   |   |
| N | 2 |   |   |   |   |   |   |   |   |   |
| I | 1 |   |   |   |   |   |   |   |   |   |
| # | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| # |   | E | X | E | C | U | T | I | O | N |

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } S_1(i) \neq S_2(j) \\ 0; & \text{if } S_1(i) = S_2(j) \end{cases} \end{cases}$$





## The Edit Distance Table

|   |   |   |   |    |    |    |    |    |    |    |
|---|---|---|---|----|----|----|----|----|----|----|
| N | 9 | 8 | 9 | 10 | 11 | 12 | 11 | 10 | 9  | 8  |
| O | 8 | 7 | 8 | 9  | 10 | 11 | 10 | 9  | 8  | 9  |
| I | 7 | 6 | 7 | 8  | 9  | 10 | 9  | 8  | 9  | 10 |
| T | 6 | 5 | 6 | 7  | 8  | 9  | 8  | 9  | 10 | 11 |
| N | 5 | 4 | 5 | 6  | 7  | 8  | 9  | 10 | 11 | 10 |
| E | 4 | 3 | 4 | 5  | 6  | 7  | 8  | 9  | 10 | 9  |
| T | 3 | 4 | 5 | 6  | 7  | 8  | 7  | 8  | 9  | 8  |
| N | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 7  | 8  | 7  |
| I | 1 | 2 | 3 | 4  | 5  | 6  | 7  | 6  | 7  | 8  |
| # | 0 | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
|   | # | E | X | E  | C  | U  | T  | I  | O  | N  |



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## Computing alignments

➤ Edit distance isn't sufficient

▶ We often need to **align** each character of the two strings to each other

➤ We do this by keeping a "backtrace"

➤ Every time we enter a cell, remember where we came from

➤ When we reach the end,

▶ Trace back the path from the upper right corner to read off the alignment



## The Edit Distance Table

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| N | 9 |   |   |   |   |   |   |   |   |   |
| O | 8 |   |   |   |   |   |   |   |   |   |
| I | 7 |   |   |   |   |   |   |   |   |   |
| T | 6 |   |   |   |   |   |   |   |   |   |
| N | 5 |   |   |   |   |   |   |   |   |   |
| E | 4 |   |   |   |   |   |   |   |   |   |
| T | 3 |   |   |   |   |   |   |   |   |   |
| N | 2 |   |   |   |   |   |   |   |   |   |
| I | 1 |   |   |   |   |   |   |   |   |   |
| # | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|   | # | E | X | E | C | U | T | I | O | N |

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } S_1(i) \neq S_2(j) \\ 0; & \text{if } S_1(i) = S_2(j) \end{cases} \end{cases}$$



## MinEdit with Backtrace

|          |          |          |          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| <b>n</b> | 9        | ↓ 8      | ↙↘ 9     | ↙↘ 10    | ↙↘ 11    | ↙↘ 12    | ↓ 11     | ↓ 10     | ↓ 9      | ↙ 8      |
| <b>o</b> | 8        | ↓ 7      | ↙↘ 8     | ↙↘ 9     | ↙↘ 10    | ↙↘ 11    | ↓ 10     | ↓ 9      | ↙ 8      | ← 9      |
| <b>i</b> | 7        | ↓ 6      | ↙↘ 7     | ↙↘ 8     | ↙↘ 9     | ↙↘ 10    | ↓ 9      | ↙ 8      | ← 9      | ← 10     |
| <b>t</b> | 6        | ↓ 5      | ↙↘ 6     | ↙↘ 7     | ↙↘ 8     | ↙↘ 9     | ↙ 8      | ← 9      | ← 10     | ↙↘ 11    |
| <b>n</b> | 5        | ↓ 4      | ↙↘ 5     | ↙↘ 6     | ↙↘ 7     | ↙↘ 8     | ↙↘ 9     | ↙↘ 10    | ↙↘ 11    | ↙↘ 10    |
| <b>e</b> | 4        | ↙ 3      | ← 4      | ↙↘ 5     | ← 6      | ← 7      | ↙↘ 8     | ↙↘ 9     | ↙↘ 10    | ↓ 9      |
| <b>t</b> | 3        | ↙↘ 4     | ↙↘ 5     | ↙↘ 6     | ↙↘ 7     | ↙↘ 8     | ↙ 7      | ↙↘ 8     | ↙↘ 9     | ↓ 8      |
| <b>n</b> | 2        | ↙↘ 3     | ↙↘ 4     | ↙↘ 5     | ↙↘ 6     | ↙↘ 7     | ↙↘ 8     | ↓ 7      | ↙↘ 8     | ↙ 7      |
| <b>i</b> | 1        | ↙↘ 2     | ↙↘ 3     | ↙↘ 4     | ↙↘ 5     | ↙↘ 6     | ↙↘ 7     | ↙ 6      | ← 7      | ← 8      |
| <b>#</b> | 0        | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8        | 9        |
|          | <b>#</b> | <b>e</b> | <b>x</b> | <b>e</b> | <b>c</b> | <b>u</b> | <b>t</b> | <b>i</b> | <b>o</b> | <b>n</b> |



## Adding Backtrace to Minimum Edit Distance

### ➤ Base conditions:

$$D(i,0) = i \quad D(0,j) = j$$

### ➤ Termination:

$D(N,M)$  is distance

### ➤ Recurrence Relation:

For each  $i = 1 \dots M$

For each  $j = 1 \dots N$

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 & \text{deletion} \\ D(i,j-1) + 1 & \text{insertion} \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } X(i) \neq Y(j) \\ 0; & \text{if } X(i) = Y(j) \end{cases} & \text{substitution} \end{cases}$$

$$\text{ptr}(i,j) = \begin{cases} \text{LEFT} & \text{insertion} \\ \text{DOWN} & \text{deletion} \\ \text{DIAG} & \text{substitution} \end{cases}$$



## Result of Backtrace

➤ Two strings and their **alignment**:

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| I | N | T | E | * | N | T | I | O | N |
|   |   |   |   |   |   |   |   |   |   |
| * | E | X | E | C | U | T | I | O | N |



➤ Time:

$O(nm)$

➤ Space:

$O(nm)$

➤ Backtrace

$O(n+m)$





## Much More Space to Work

- Contextual information
- Language models
- Sources of errors
- Domain knowledge
- Common mistakes



## Further Reading

➤ Speech and Language Processing (3<sup>rd</sup> ed. draft)

▶ Chapter 2